Experiment 8 : Architecture Diagram

Steps to create diagram

1. Open draw.io (diagrams.net)
2. Create Blank Diagram
3. Add layered containers:
   User Layer ,Frontend Layer,Backend Layer, Data Layer
4. Add components:

- User Layer:User
- Frontend Layer: Text Overlay (cv2.putText) ,Audio Output (pyttsx3),OpenCV Display (cv2.imshow)
- Backend Layer:Webcam (Camera Input),OpenCV (Frame Capture),MediaPipe Hands (21 landmarks)
  Feature Vector (42 values),Random Forest Model (15 classes),Prediction Logic (confidence > 0.80)
- Data Layer:Dataset Folder (training only),model.pkl (trained model)

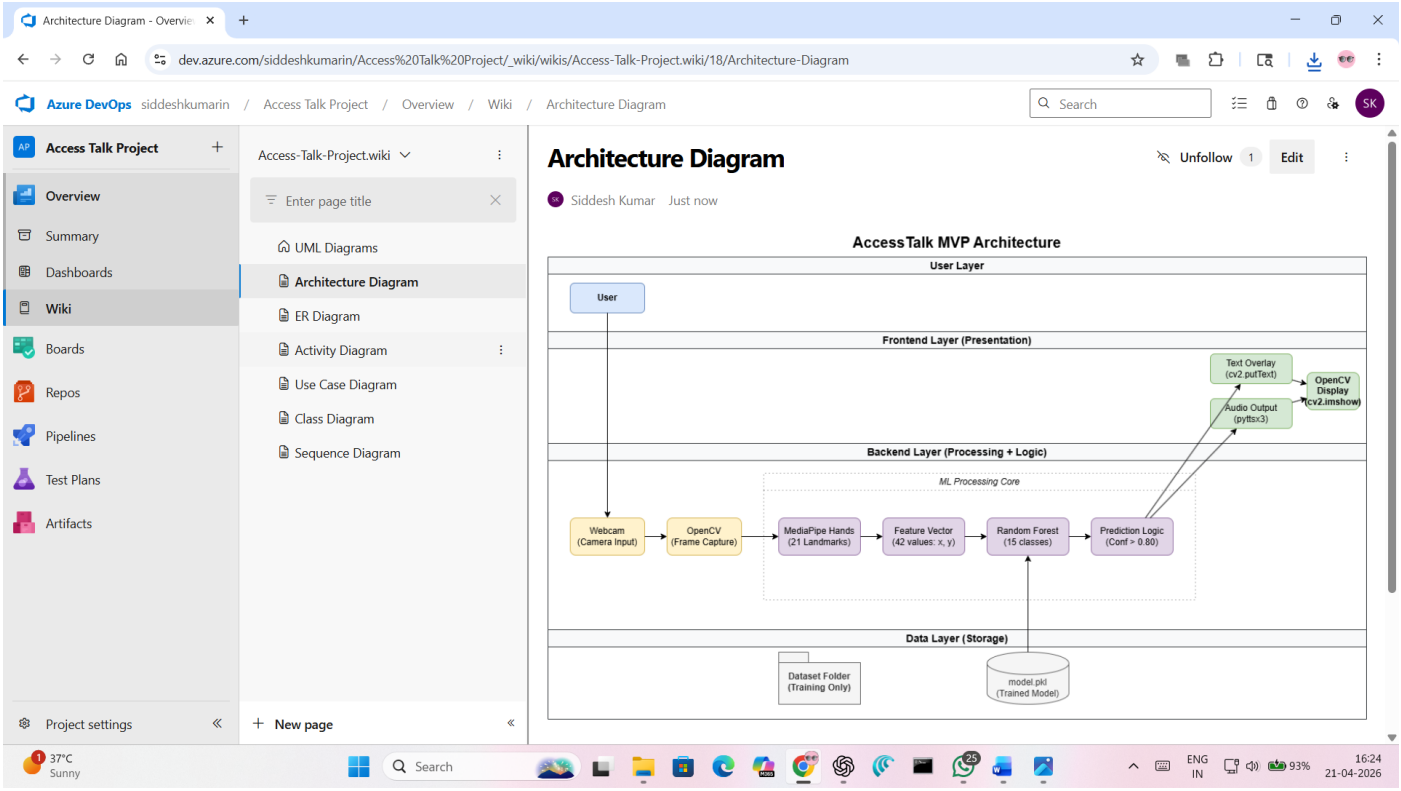
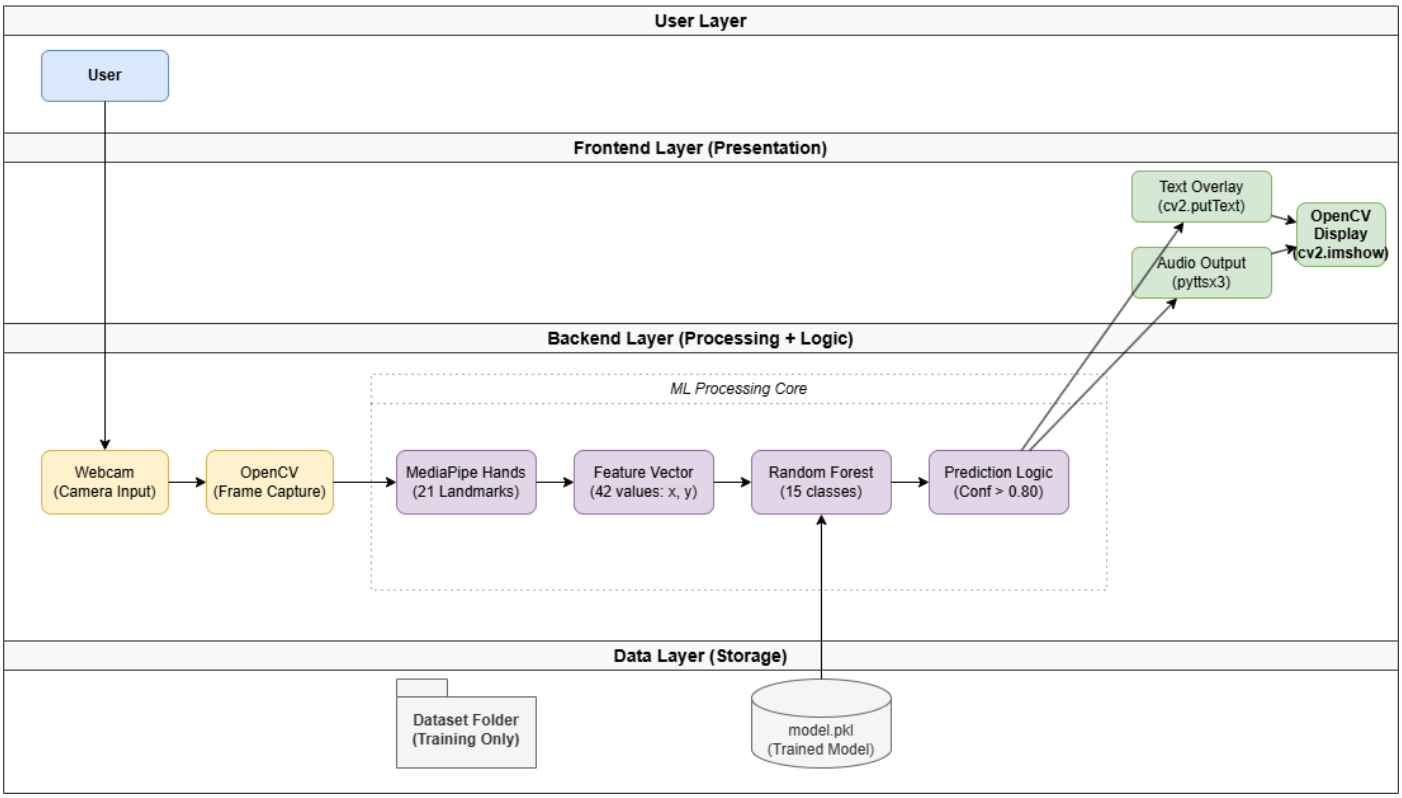
5. Connect arrows
6. Arrange layout:
   Top: User Layer
   Middle: Frontend + Backend
   Bottom: Data Layer

Diagram Notations

• User Layer → interaction layer
• Frontend → UI (display + speech output)
• Backend → processing + ML pipeline
• Data Layer → storage (model + dataset)

• Webcam → input source
• MediaPipe → landmark detection
• Random Forest → classification model
• Prediction Logic → confidence filtering

• model.pkl → trained knowledge (acts like database)
• Dataset → training data (not used in runtime)



AccessTalk MVP Architecture